

Problem-1

Direct block address = 8

Single indirect blocks = 2

Double " " = 2

Triple " " = 2

Size of Block = 32B

(Size of block address) = 4B

Soln:

1 block can store $= \frac{32}{4} = 8$ addresses

↓
(Single indirect)

Total block addresses stored

$$= 8 + (2 \times 8) + (2 \times 8^2) + (2 \times 8^3)$$

$$= (8 + 16 + 128 + 1024)$$

$$= 1176$$

$$\therefore \text{Max possible file size} = (1176 \times 32) \text{ B}$$

$$= 37632 \text{ B}$$

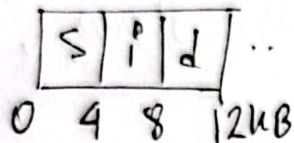
$$= 36.75 \text{ KB}$$

(Ans)

Problem-2

Inode size = 512 B

Block size = 4 KB



Inode number = 23

size of sector = 256 B

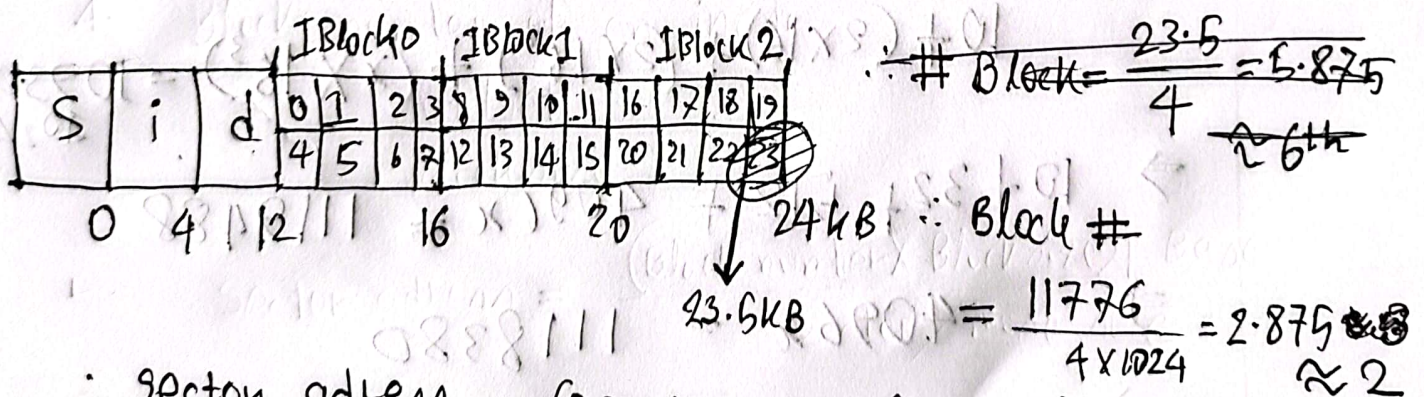
Soln:

Base address of inode = $4 \times 3 = 12 \text{ KB}$

offset of inode # 23 = $23 \times 512 \text{ B} = 11776 \text{ B}$

Starting

$\therefore$ address of inode # 23 = $12 + 11.5 = 23.5 \text{ KB}$



$\therefore$ sector address

$$= \frac{(2.875 \times 4 \times 1024) + (12 \times 1024)}{256} = \boxed{94}$$

Spring 25 (set A)

[03]

b.

direct block address = 10

Single indirect blocks = 2

Double indirect blocks = 2

size of a block = 128B

size of address = 8B

Soln:

1 indirect block can store = $\frac{128}{8} = 16$ Block addresses

Let's assume

of triple indirect block = x

Now,

$$10 + (2 \times 16) + (2 \times 16^2) + (x \times 16^3) = 1093.25 \times 1024$$

$$\Rightarrow 10 + 32 + 512 + 4096x = 1119488$$

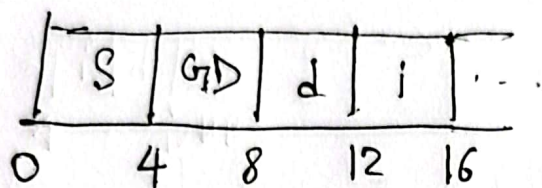
$$\Rightarrow 4096x = 1118380$$

$$\Rightarrow x = 273.177 \approx \boxed{274}$$

c.

$$\text{inode size} = 128 \text{ B}$$

$$\text{Block size} = 4 \text{ KB}$$



$$\text{inode number} = 84$$

$$\text{sector size} = 1 \text{ KB}$$

Soln:

$$\therefore \text{Inode address} = (84 \times 128) + (16 \times 1024)$$
$$= 27136 \text{ B}$$

$$= \boxed{26.5 \text{ KB}}$$

$$\therefore \text{Inode Block number} = \frac{84 \times 128}{4 \times 1024} = 2.625 \approx 2$$

$$\therefore \text{Sector address} = \frac{(2.625 \times 1024 \times 4) + (16 \times 1024)}{1024}$$

$$= 26.5 \text{ KB} = \boxed{27136 \text{ B}}$$